

Lab 6

Oops

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Dept: Robotics and Ai

Task1:

```
#include <iostream>
using namespace std;
struct Book
{
    int id;
    float price;
    int pages;
};

int main()
{
    Book b1, b2, b3;

    cout << "Enter details for Book 1\n";
    cout << "ID: ";
    cin >> b1.id;
    cout << "Price: ";
    cin >> b1.price;
    cout << "Pages: ";
    cin >> b1.pages;

    cout << "\nEnter details for Book 2\n";
    cout << "ID: ";
    cin >> b2.id;
    cout << "Price: ";
    cin >> b2.price;
    cout << "Pages: ";
    cin >> b2.pages;

    cout << "\nEnter details for Book 3\n";
    cout << "ID: ";
    cin >> b3.id;
    cout << "Price: ";
```

```
Enter details for Book 1
ID: 56
Price: 2500
Pages: 100

Enter details for Book 2
ID: 34
Price: 1500
Pages: 99

Enter details for Book 3
ID: 20089
Price: 3000
Pages: 101

Enter 1 to show most costly book
Enter 2 to re-enter details
1

Most Costly Book:
ID: 20089
Price: 3000
Pages: 101

-----
Process exited after 72.32 seconds with return value 0
Press any key to continue . . . |
```

Task2

```

/*tasr2
#include <iostream>
#include <string>
using namespace std;

struct Citizen
{
    string name;
    string cnic;
    string city;
};

int main()
{
    Citizen c;
    Citizen *ptr = &c;

    cout << "Enter Name: ";
    getline(cin, ptr->name);

    cout << "Enter CNIC: ";
    getline(cin, ptr->cnic);

    cout << "Enter City: ";
    getline(cin, ptr->city);

    int choice;
    cout << "\nPress 1 to update city, otherwise press any number: ";
    cin >> choice;

```

```

Enter Name: Qadeer
Enter CNIC: 35302333334
Enter City: Lahore

Press 1 to update city, otherwise press any number: 1
Enter New City: karachi

Updated Record:
Name: Qadeer
CNIC: 35302333334
City: karachi

-----
Process exited after 46.31 seconds with return value 0
Press any key to continue . . . |

```

Task 3

```

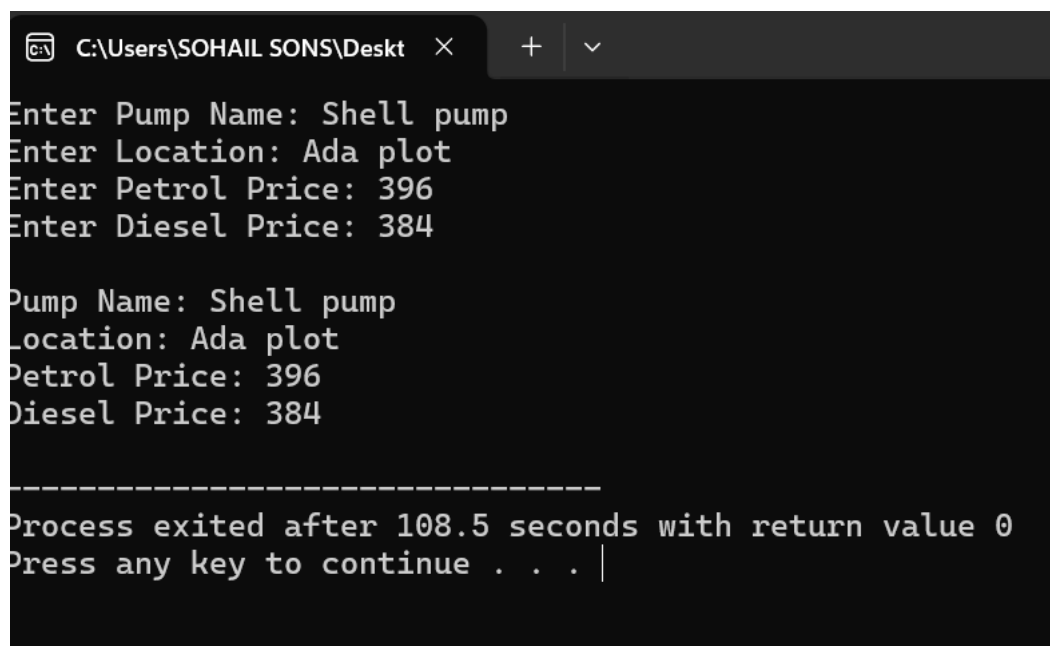
/*task3
#include <iostream>
#include <string>
using namespace std;

struct PetrolPump{
    string name;
    string location;
    float petrolPrice;
    float dieselPrice;
};

void inputData(PetrolPump &p){
    cout<<"Enter Pump Name: ";
    getline(cin,p.name);
    cout<<"Enter Location: ";
    getline(cin,p.location);
    cout<<"Enter Petrol Price: ";
    cin>>p.petrolPrice;
    cout<<"Enter Diesel Price: ";
    cin>>p.dieselPrice;
    cin.ignore();
}

void displayData(PetrolPump p){
    cout<<"\nPump Name: "<<p.name;
    cout<<"\nLocation: "<<p.location;
}

```



```

C:\Users\SOHAIL SONS\Desktop
Enter Pump Name: Shell pump
Enter Location: Ada plot
Enter Petrol Price: 396
Enter Diesel Price: 384

Pump Name: Shell pump
Location: Ada plot
Petrol Price: 396
Diesel Price: 384

-----
Process exited after 108.5 seconds with return value 0
Press any key to continue . . . |

```

Task 4

```

/*task4
#include <iostream>
#include <string>
using namespace std;

struct Bill{
string name;
int units;
float rate;
float total;
};

void inputBill(Bill &b){
cout<<"Enter Name: ";
getline(cin,b.name);
cout<<"Enter Units: ";
cin>>b.units;
cout<<"Enter Rate per Unit: ";
cin>>b.rate;
cin.ignore();
}

void calculateBill(Bill &b){
b.total=b.units*b.rate;
}

void displayBill(Bill b){
cout<<"\n--- Electricity Bill ---";
cout<<"\nName: "<<b.name;
cout<<"\nUnits: "<<b.units;
cout<<"\nRate: "<<b.rate;
cout<<"\nTotal Bill: "<<b.total<<endl;
}

```

```

Enter Name: Qadeer
Enter Units: 5
Enter Rate per Unit: 199

```

```

--- Electricity Bill ---
Name: Qadeer
Units: 5
Rate: 199
Total Bill: 995

```

```

-----
Process exited after 37.94 seconds with return value 0
Press any key to continue . . . |

```

Home task

Task1

```
#include <string>
using namespace std;

struct Patient
{
    int id;
    string name;
    float temperature;
    int severity;
};

Patient* getCritical(Patient p[],int n)
{
    Patient* maxPtr=&p[0];
    for(int i=1;i<n;i++)
    {
        if(p[i].severity>maxPtr->severity)
        {
            maxPtr=&p[i];
        }
    }
    return maxPtr;
}

void showCritical(Patient p[],int n)
{
    Patient* maxPtr=getCritical(p,n);
    cout<<"\nCritical Patient(s)\n";
    for(int i=0;i<n;i++)
    {
        if(p[i].severity==maxPtr->severity)
```

```
Again(y/n) :
Enter ID:Not found

Critical Patient(s)
Again(y/n) :
Ente
```